

ESP32-C3 Cyber Keychain

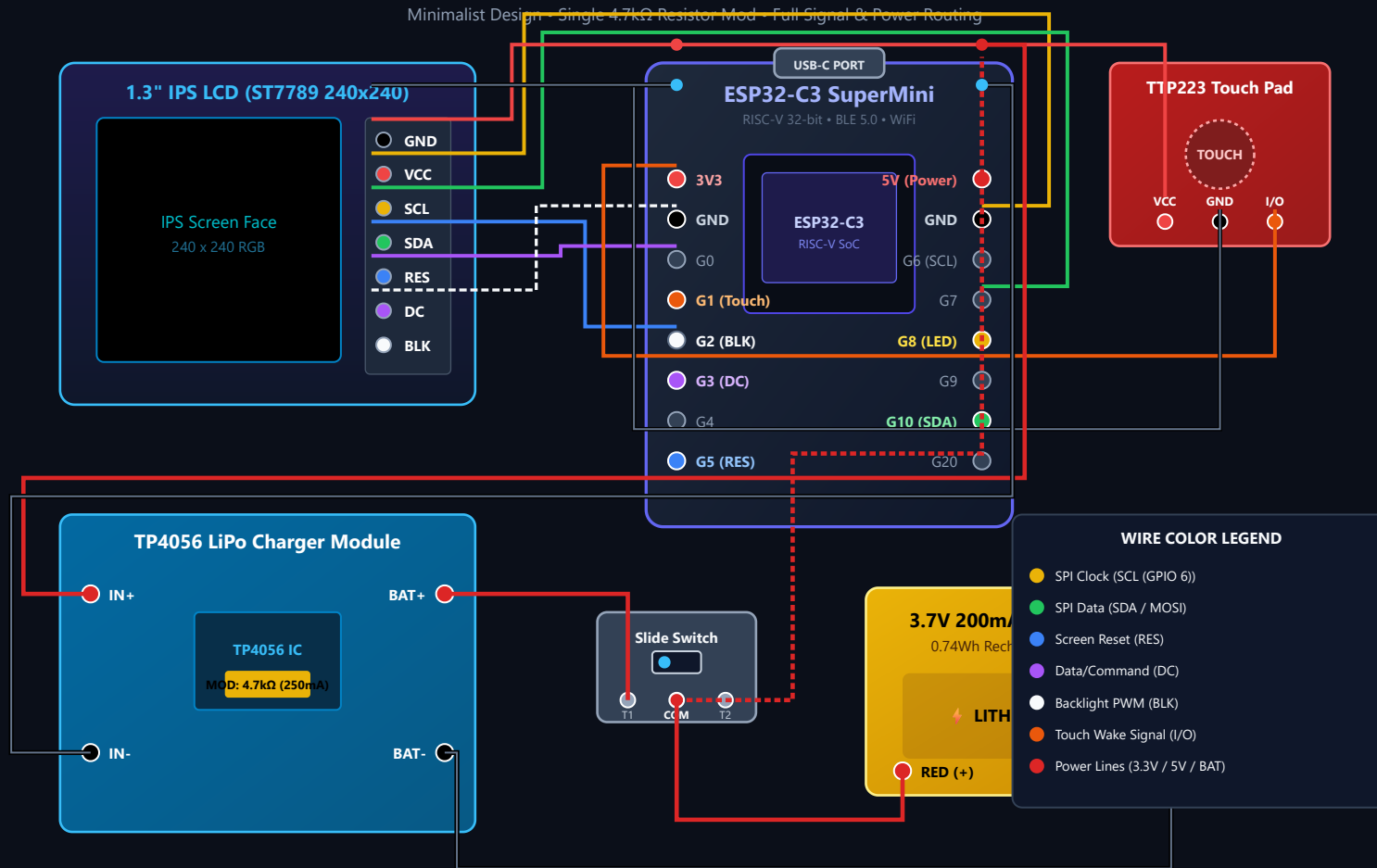
Master Circuit & Hardware Wiring Manual

Visual Diagrams • Zero-Clutter Routing • Step-by-Step Soldering



Complete Visual Circuit Blueprint

⚡ COMPLETE ESP32-C3 CYBER KEYCHAIN WIRING BLUEPRINT



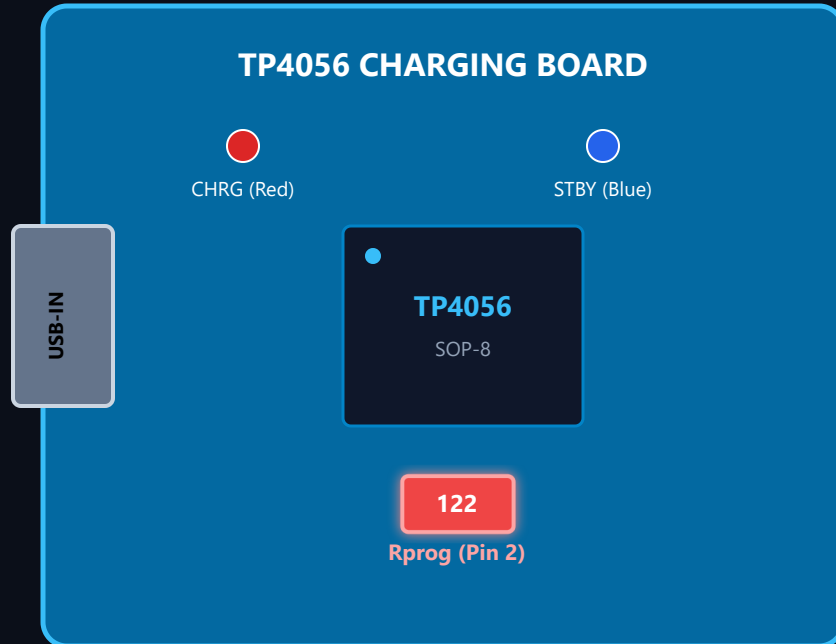
Pin-by-Pin Master Connection Chart

Module & Pin	Connects to ESP32 Pin	Wire Color	Electrical Function & Signal
Display GND	ESP32 GND	● Black	Screen Power Ground
Display VCC	ESP32 3.3V	● Red	3.3V Regulated Power Rail
Display SCL	ESP32 GPIO 8	● Yellow	Hardware SPI Clock (SCK @ 40MHz)
Display SDA	ESP32 GPIO 10	● Green	Hardware SPI Data (MOSI)
Display RES	ESP32 GPIO 5	● Blue	Screen Hardware Reset Signal
Display DC	ESP32 GPIO 3	● Purple	Data (High) / Command (Low) Select
Display BLK	ESP32 GPIO 2	● White	Backlight PWM Dimming (0–255)
Touch VCC	ESP32 3.3V	● Red	Touch Sensor 3.3V Power
Touch GND	ESP32 GND	● Black	Touch Sensor Ground
Touch I/O	ESP32 GPIO 1	● Orange	Deep Sleep RTC Wakeup Interrupt

The TP4056 Resistor Mod (Illustrated)

TP4056 CURRENT MODIFICATION FOR 200mAh BATTERY

Step-by-step: Swap 1.2k Ω (122) SMD Resistor with 4.7k Ω for Safe 250mA Charging



The Charging Formula:

$$I_{\text{charge}} = 1200 / R_{\text{prog}}$$

✗ Stock Resistor (122 = 1.2k Ω):

$$I = 1200 / 1200 = \mathbf{1000mA (1A)}$$

Too high! Will puff up a 200mAh LiPo.

✓ Swapped Resistor (4.7k Ω):

$$I = 1200 / 4700 = \mathbf{\sim 255mA (0.25A)}$$

Safe, gentle charging in ~50 minutes!

Tip: Heat with soldering iron tip,
flick off '122' & solder 4.7k Ω across pads!

Power, Charging & Switch Flow Explained

How the Single USB-C Port Powers & Charges Everything:

1. When USB-C is Plugged In:

- The ESP32-C3 SuperMini's USB-C port receives 5V.
- Power leaves the ESP32's 5V pin and enters the TP4056's IN+ pad.
- The TP4056 steps down and regulates charging at 255mA safely into the 200mAh LiPo through BAT+ and BAT-.

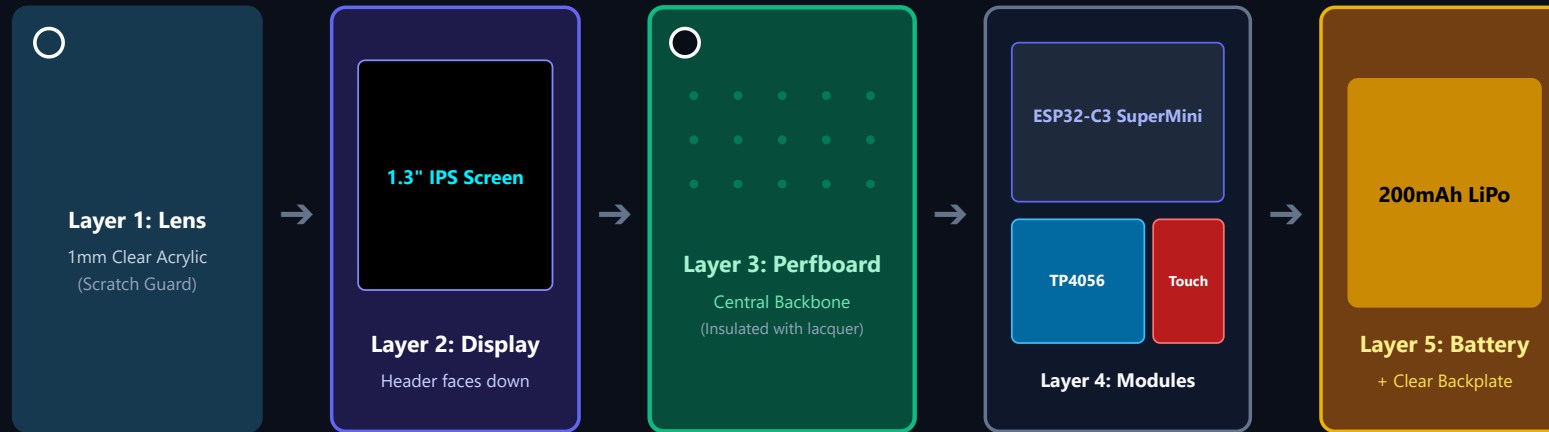
2. When Running on Battery (Unplugged):

- Battery outputs 3.7V to 4.2V.
- Current passes through the SPDT Slide Switch into the ESP32's 5V pin.
- The ESP32-C3 SuperMini's onboard 3.3V LDO regulator steps it down to a stable, clean 3.3V to feed the chip, the 1.3" display, and the touch sensor!

🍰 Physical Perfboard Sandwich Assembly

PHYSICAL PERFBORARD SANDWICH ASSEMBLY

Exploded View • 35mm x 55mm Form Factor • Rugged Backpack Construction



🔧 Sandwich Assembly Pro-Tips for Maximum Durability:

1. **Cut Perfboard First:** Cut green perfboard to ~35mm x 55mm so it has a 5mm margin around the 1.3" display.
2. **Keyring Hole:** Drill a 3.5mm hole in the top corner of the perfboard BEFORE soldering components.
3. **Foam Tape Cushion:** Stick the 200mAh battery behind the ESP32 using double-sided foam tape to absorb drops.
4. **Clear Conformal Coating:** Paint solder joints with clear nail polish to make it 100% immune to backpack short-circuits.

Step-by-Step Soldering Sequence (1 to 10)

Follow this order to prevent damaging heat-sensitive parts:

1. **Step 1 (Cut & Drill):** Cut perfboard to $\approx 35\text{mm} \times 55\text{mm}$. Drill a 3.5mm keyring hole in the top corner.
2. **Step 2 (TP4056 Mod):** Desolder SMD resistor 122 (1.2k Ω) on the TP4056 and solder your 4.7k Ω **resistor** across the pads.
3. **Step 3 (Mount Display):** Solder the 7-pin header of the 1.3" IPS LCD on the front face of the perfboard.
4. **Step 4 (Mount ESP32):** Solder the ESP32-C3 SuperMini on the back face of the perfboard.
5. **Step 5 (Wire SPI Data Lines):** Solder Yellow (GPIO8), Green (GPIO10), Blue (GPIO5), Purple (GPIO3), and White (GPIO2) from display to ESP32.
6. **Step 6 (Wire 3.3V & GND Rails):** Connect 3.3V and GND to Display and TTP223 Touch sensor.
7. **Step 7 (Mount Switch):** Solder the 4mm SPDT Slide Switch onto the side edge of the perfboard.
8. **Step 8 (Wire TP4056 Charger):** Wire ESP32 5V $\rightarrow$ IN+, ESP32 GND $\rightarrow$ IN-.
9. **Step 9 (Connect Battery):** Solder Battery Black (-) to BAT-, and Battery Red (+) through the switch to ESP32 5V.
10. **Step 10 (Insulate):** Paint all solder joints on the back with clear nail polish to prevent short-circuits.

Pre-Power Safety Verification Checklist

Before sliding the switch to "ON", check these with a Multimeter (Continuity Mode):

- ☐ **Ground Continuity Test:** Probe ESP32 `GND`, Display `GND`, TP4056 `IN-`, and Battery `(-)` — all must beep together (shared ground).
- ☐ **No 3.3V to GND Short:** Probe `3.3V` and `GND` — must **NOT** beep (infinite or high resistance).
- ☐ **No 5V to GND Short:** Probe `5V` and `GND` — must **NOT** beep.
- ☐ **Resistor Mod Verified:** Measure resistance across TP4056 Pin 2 and GND — should read $\approx 4.7\text{k}\Omega$.
- ☐ **Polarity Check:** Ensure Red wire from LiPo goes to `BAT+` / Switch, and Black wire goes to `BAT-` / `GND`.



Troubleshooting Matrix

Symptom	Probable Cause	Exact Fix
Screen is completely black	Backlight BLK pin is floating	Ensure BLK wire is soldered securely to GPIO 2 .
Screen has power but no image	SPI Clock or Data pins swapped	Swap SCL (must be GPIO8) and SDA (must be GPIO10).
Touch sensor doesn't wake device	Wrong GPIO or mode	Ensure touch I/O wire is on GPIO 1 (RTC GPIO domain).
ESP32 reboots continuously	Battery voltage is < 3.0V	Plug in USB-C cable for 30 minutes to charge the battery.
Battery gets hot while charging	Stock 1A resistor still installed	Verify resistor 122 was replaced with 4.7kΩ.

Your Wiring is 100% Complete!

-  **Printable Circuit PDF:** `wiring_guide.pdf`
-  **Interactive Circuit Slides:** `wiring_guide.html`
-  **High-Res Vector Diagrams:** Inside `assets/` directory

⚡ Happy Building & Soldering! ⚡